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DEPARTMENT OF CSE (Artificial Intelligence)

Assignment –AI QUESTION GENERATOR

2025-2026

Subject Name: Agentic AI Systems

Subject Code: M23BCA602

Semester: 6th

Submitted by: Team No:

Sl. No.	Student Name	USN.		CO's Mapping				Total	Scaled to
			CO1	CO2	CO3	CO4	CO5		
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ABSTRACT

Internships are an essential component of higher education as they provide students with practical exposure to real-world work environments and help bridge the gap between academic learning and industry requirements. Through internships, students gain hands-on experience, develop technical and professional skills, and understand workplace practices. However, in many educational institutions, internship monitoring and evaluation are still carried out using manual methods such as spreadsheets, paper-based reports, and periodic submissions. These traditional approaches often result in inefficient tracking of student progress, difficulty in assessing learning outcomes, lack of timely feedback, and increased administrative burden on both students and mentors.

To address these challenges, this project proposes an Agentic Internship Progress Tracker and Reflection Assistant, an intelligent AI-powered system designed to automate internship monitoring, performance analysis, and reflection generation. The system leverages the principles of Agentic Artificial Intelligence, where autonomous agents perform specific tasks, utilize multiple tools, maintain memory of previous interactions, and continuously evaluate their outputs to provide meaningful assistance. Unlike conventional internship management systems, the proposed solution not only records internship activities but also analyzes student progress, identifies pending tasks, extracts acquired skills, generates reflective summaries, and assists in preparing final internship reports.

The system incorporates multiple intelligent agents, including a Weekly Reflection Agent, Skill Extraction Agent, Pending Task Detection Agent, and Report Generation Agent. These agents work collaboratively to process internship data submitted by students on a weekly basis. The Reflection Agent analyzes completed tasks, challenges faced, and learning experiences to generate structured reflections and personalized improvement suggestions. The Skill Extraction Agent identifies both technical and soft skills developed during the internship, helping students understand their growth and competencies. The Pending Task Detection Agent compares planned objectives with completed work to identify delays and unfinished activities, enabling better time management and accountability. Additionally, the Report Generation Agent compiles internship records and automatically creates comprehensive reports, reducing the effort required during final submissions.

A key feature of the proposed system is its memory and state management capability. Internship activities, reflections, mentor feedback, and progress history are stored in a database, allowing the agent to track development over time and provide context-aware recommendations. The system also incorporates a reflection and replanning mechanism, where the agent reviews its generated outputs, detects missing information, and requests clarification when necessary. This self-evaluation capability enhances the accuracy and usefulness of the generated insights.

The application is developed using a modern technology stack consisting of a user-friendly web interface, backend processing services, database storage, and AI integration through large language models. Students can easily submit weekly logs, track their learning progress, and access AI-generated reflections through the dashboard. Mentors can monitor student performance, review analytical reports, and provide feedback using a dedicated mentor interface. The system offers data visualization and progress analytics to facilitate informed decision-making and effective internship management.

1.INTRODUCTION

Internships have become an integral part of modern education as they provide students with valuable opportunities to apply theoretical knowledge in real-world professional environments. Through internships, students gain practical experience, develop technical and interpersonal skills, understand workplace culture, and improve their employability. Educational institutions increasingly encourage students to participate in internships because they serve as a bridge between academic learning and industry requirements. However, despite their importance, the process of tracking and evaluating internship progress remains largely manual and inefficient in many organizations and institutions.

Traditionally, internship monitoring relies on weekly reports, spreadsheets, handwritten logs, email communications, and periodic mentor evaluations. Students are often required to maintain records of their activities and submit progress reports manually. Mentors and faculty coordinators must review these reports individually, making the process time-consuming and difficult to manage, especially when handling large numbers of interns. Furthermore, traditional systems focus primarily on recording activities rather than analyzing learning outcomes, identifying challenges, or providing personalized feedback. As a result, students may fail to recognize areas for improvement, while mentors may struggle to obtain a clear understanding of each student's progress and achievements.

Another significant challenge in internship management is the lack of continuous monitoring and intelligent assessment. Students frequently forget to document important tasks, overlook learning experiences, or provide incomplete reports. Similarly, mentors may not receive timely information about delays, pending tasks, or difficulties faced by interns. Without an effective mechanism to track progress and analyze performance, opportunities for guidance and improvement are often missed. This limitation reduces the overall effectiveness of internship programs and makes it difficult to measure their educational impact accurately.

Recent advancements in Artificial Intelligence (AI), particularly Agentic AI systems, provide an opportunity to transform the way internships are monitored and managed. Agentic AI refers to intelligent systems that can autonomously perform tasks, utilize multiple tools, maintain memory of previous interactions, make decisions based on available information, and continuously evaluate their outputs to achieve specific goals. Unlike traditional software applications that simply store and display data, agentic systems actively analyze information, generate insights, and assist users in decision-making processes.

2.PROBLEM STATEMENT

Internships are an important part of a student's academic and professional development, providing opportunities to gain practical experience and apply theoretical knowledge in real-world environments. However, the process of monitoring and evaluating internship activities is often inefficient due to the use of traditional and manual tracking methods. Most educational institutions and organizations rely on spreadsheets, handwritten reports, emails, or static forms to record internship progress. These methods focus primarily on data collection rather than meaningful analysis and feedback.

One of the major challenges in existing internship management systems is the lack of continuous progress tracking. Students generally submit reports at fixed intervals, making it difficult for mentors and faculty coordinators to monitor ongoing activities and identify issues in a timely manner. As a result, delays, incomplete tasks, and performance concerns often remain unnoticed until the end of the internship period.

Another significant problem is the manual and repetitive nature of report preparation. Students are required to create weekly reports, reflections, and final summaries, which can be time-consuming and prone to inconsistencies. Similarly, mentors must spend considerable effort reviewing these reports individually, increasing their administrative workload and reducing the time available for meaningful guidance and mentorship.

Existing systems also struggle to identify pending tasks and project delays effectively. Since progress updates are often stored as unstructured text, there is no intelligent mechanism to compare planned objectives with completed work. Consequently, students may fall behind schedule without receiving timely alerts or recommendations for corrective action.

Current internship monitoring systems face several challenges:

- Lack of continuous progress tracking.
- Manual and repetitive report preparation.
- Difficulty in identifying pending tasks and delays.
- Limited mentor visibility into student performance.
- Absence of structured reflection and learning assessment.
- No intelligent mechanism to analyze growth and skill development.

These issues reduce the effectiveness of internships and increase administrative workload for students and mentors.

3.OBJECTIVES

The primary objective of the proposed Agentic Internship Progress Tracker and Reflection Assistant is to improve the efficiency and effectiveness of internship monitoring by integrating Artificial Intelligence, automation, and intelligent decision-making capabilities. The system is designed to support students, mentors, and educational institutions by providing continuous progress tracking, reflective learning support, and automated performance analysis. The specific objectives of the project are as follows:

The main objectives of the proposed system are:

1. To Automate Weekly Internship Progress Trac king

The system aims to eliminate manual tracking methods by providing a digital platform where students can regularly submit internship activities, completed tasks, work hours, challenges faced, and future goals. Automated tracking ensures that internship progress is recorded consistently and accurately throughout the internship duration.

2. To Generate AI-Based Reflections and Learning Summaries

The proposed system utilizes Artificial Intelligence to analyze internship activities and generate structured reflections. These reflections help students understand their learning experiences, evaluate their performance, identify achievements, and recognize areas that require improvement. The generated summaries encourage continuous learning and professional development.

3. To Identify Pending and Incomplete Tasks Automatically

A key objective of the system is to compare planned objectives with completed activities and identify unfinished work. The agent continuously monitors task completion status and generates alerts for pending assignments or delayed activities, enabling students to manage their responsibilities more effectively.

4. To Extract Technical and Professional Skills from Internship Activities

The system automatically analyzes submitted tasks and identifies technical skills, tools, and professional competencies acquired during the internship. This feature helps students track their skill development and provides mentors with measurable indicators of learning outcomes and professional growth.

5. To Maintain Internship History Using Memory and State Management

project incorporates memory management mechanisms to store internship records, weekly logs, reflections, mentor feedback, and performance history. By maintaining historical data, the system can provide context-aware recommendations, monitor long-term progress, and generate comprehensive reports based on accumulated internship experiences.

6. To Generate Final Internship Reports Automatically

internship reports manually can be time-consuming and repetitive. Therefore, the system aims to automate report generation by compiling weekly logs, reflections, skills **acquired, achievements, and mentor feedback into a structured and professional report**

4. SYSTEM ARCHITECTURE

The system consists of the following major components:

User Interface Layer

- Student Dashboard
- Mentor Dashboard
- Weekly Log Submission Portal
- Analytics Dashboard

Agent Layer

- Weekly Reflection Agent
- Skill Extraction Agent
- Pending Task Detection Agent
- Report Generation Agent
- Memory Management Agent

Tool Layer

- Log Analyzer
- Skill Mapper
- Reflection Generator
- Report Generator
- Progress Tracker

Database Layer

- SQLite/MongoDB
- User Data
- Weekly Logs
- Skills Database
- Reflection History
- Mentor Feedback

AI Layer

- Gemini API
- Prompt Processing
- Reflection Analysis

5.METHODOLOGY

The proposed system follows an agentic workflow:

Step 1: Data Collection

Students submit weekly internship details including:

- Tasks completed
- Challenges faced
- Hours worked
- Next-week goals

Step 2: Tool Execution

The agent invokes multiple tools:

- Weekly Log Analyzer
- Skill Extraction Tool
- Pending Task Detector
- Reflection Generator
- Report Generator

Step 3: Memory Storage

All internship records are stored in the database for future reference.

Step 4: Reflection and Replanning

The agent reviews generated outputs and checks for missing information. If insufficient details are detected, it requests clarification or revises the generated response.

Step 5: Output Generation

The system produces:

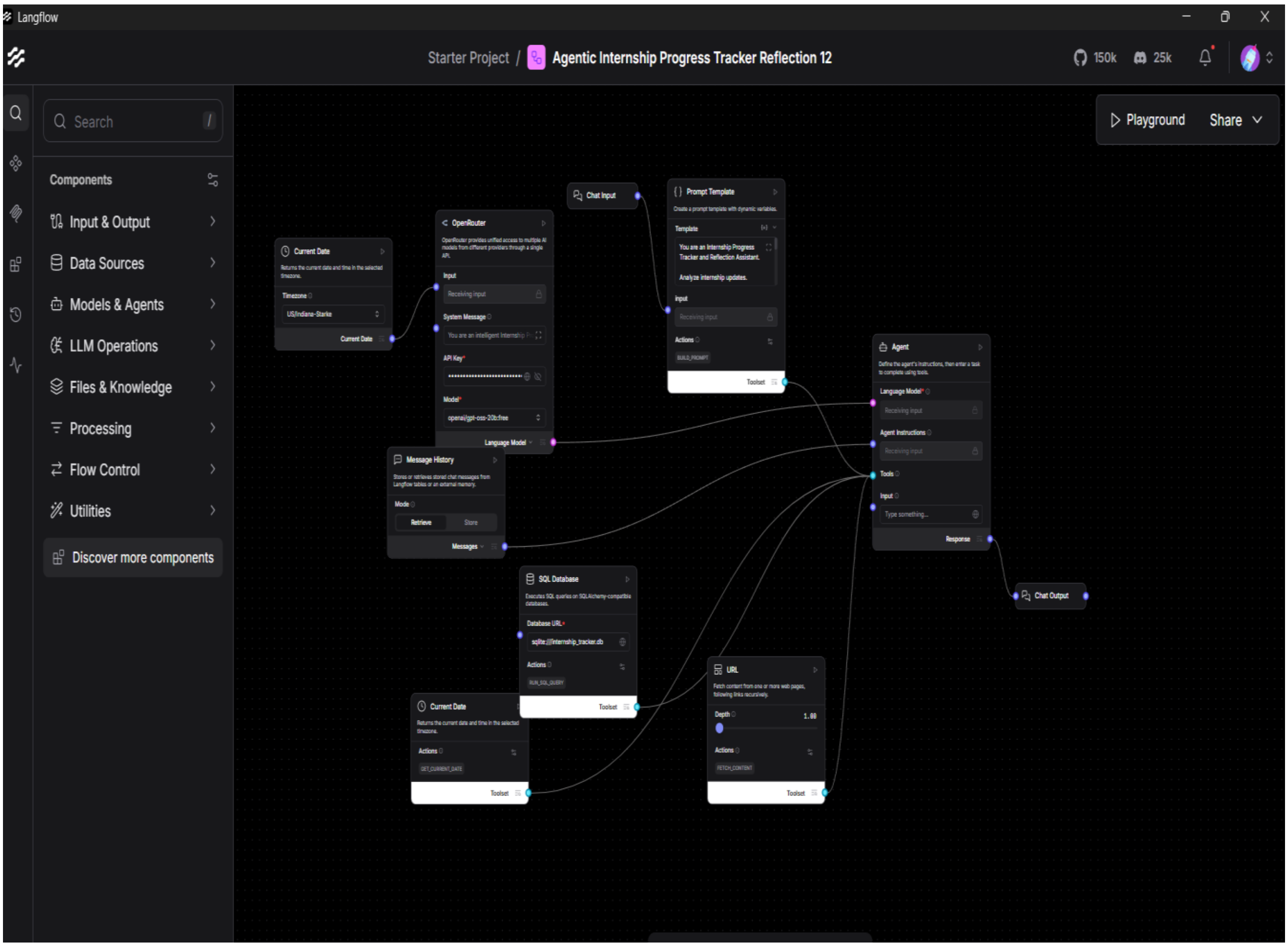
- Weekly reflections
- Skill reports
- Productivity insights
- Mentor summaries

6. IMPLEMENTATION

The proposed Agentic Internship Progress Tracker and Reflection Assistant was implemented using Langflow, an open-source visual framework for building AI-powered workflows and agent-based applications. The system integrates multiple AI components, memory mechanisms, external tools, and database connectivity to create an intelligent internship monitoring platform. The implementation focuses on providing autonomous decision-making, task analysis, memory retention, and reflection generation capabilities while maintaining a user-friendly architecture.

Tools and Technologies Used

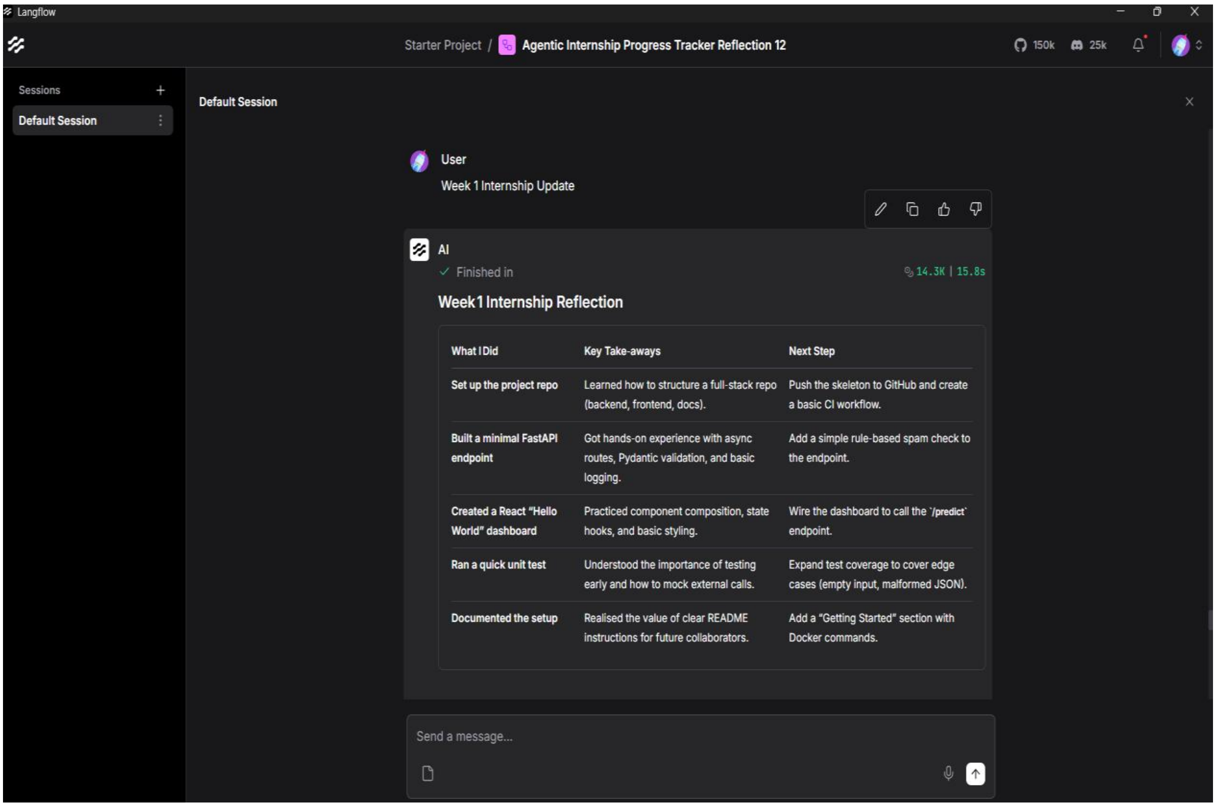
Tool / Technology	Purpose
Langflow	Visual workflow design and agent orchestration
OpenRouter API	Access to AI language models
Python	Backend processing and logic implementation
SQLite Database	Storage of internship records and memory
Prompt Engineering	Agent instruction and behavior design
Agentic AI	Multi-agent task execution and reasoning
Message History	Context retention and memory management



Workflow Components

1. Chat Input
Acts as the entry point where students submit internship updates, completed tasks, challenges, and goals.
2. Current Date Component
Provides real-time date and time information for accurate progress tracking.
3. OpenRouter Language Model
Processes user inputs and generates intelligent responses using a Large Language Model (LLM).
4. Prompt Template
Defines the agent's role, behavior, and instructions for analyzing internship activities.
5. Message History
Stores previous interactions and maintains conversational memory for personalized responses.
6. SQL Database
Stores internship records, reflections, skills, and feedback for long-term tracking.
7. URL Tool
Retrieves external information such as guidelines, resources, or reference materials when needed.
8. Agent Component
Acts as the central decision-making unit that analyzes data and generates insights, reflections, and recommendations.
9. Chat Output
Displays the final response, including progress summaries, reflections, skills identified, and suggestions.

7. RESULTS AND OUTPUT



8.FUTURE SCOPE

The proposed Agentic Internship Progress Tracker and Reflection Assistant can be further enhanced by incorporating advanced features and integrations to improve its functionality and user experience.
Future improvements may include:

- Voice-Based Reflection Submission: Allow students to submit internship reflections through voice input using speech recognition technology.
- GitHub Integration: Track coding activities, commits, repositories, and project contributions automatically from GitHub profiles.
- Jira and Trello Integration: Connect with project management tools to monitor task completion, project progress, and team collaboration activities.
- AI-Based Internship Scoring: Develop an intelligent scoring system that evaluates student performance based on task completion, consistency, learning progress, and reflection quality.
- Resume Achievement Generation: Automatically convert internship activities and acquired skills into professional resume points and career achievements.
- LinkedIn Profile Enhancement: Generate LinkedIn-ready project descriptions, skill summaries, and professional accomplishments based on internship work.
- Multi-College SaaS Platform: Expand the system into a cloud-based platform that can be used by multiple colleges and universities for internship management.
- Mobile Application Development: Develop Android and iOS applications to provide easy access to internship tracking and reporting features.
- Predictive Performance Analytics: Use machine learning techniques to predict student performance trends, identify potential risks, and recommend improvement strategies.

9. CONCLUSION

The Agentic Internship Progress Tracker and Reflection Assistant provides an effective and intelligent solution for managing and monitoring internship activities. The system automates the process of tracking student progress, generating reflections, identifying skills, detecting pending tasks, and preparing internship reports. By reducing manual work, it helps both students and mentors save time and improve productivity.

The project successfully implements key Agentic AI concepts such as tool usage, memory management, and reflection-based reasoning. The agent can store previous internship records, analyze current activities, and provide personalized recommendations based on past progress. This enables continuous monitoring and better assessment of student performance throughout the internship period.

The system also enhances learning by encouraging students to reflect on their work, identify areas for improvement, and track their skill development. Mentors benefit from detailed progress reports and analytical insights, allowing them to provide timely guidance and support.

Overall, the proposed system demonstrates the practical application of Agentic AI in internship management and educational environments. It improves efficiency, supports professional growth, and creates a more structured and data-driven internship experience. With future enhancements and additional integrations, the platform has the potential to become a comprehensive solution for internship monitoring and performance evaluation across educational institutions.